

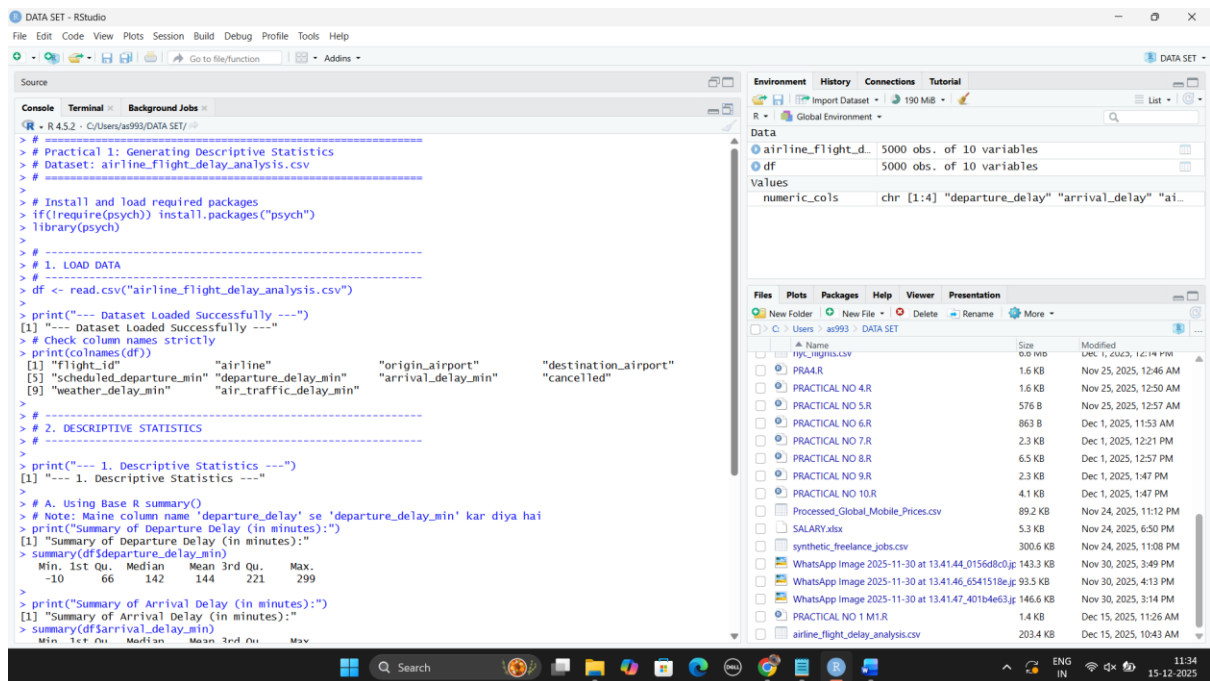
Sheth L.U.J. College of Arts And Sir M.V. College of Science and Commerce

Data Analysis with SAS / SPSS / R

PRACTICAL NO 1 MODULE 2

AIM: Generating descriptive statistics using summary() or describe()

OUTPUT:



The screenshot displays the RStudio interface. The console on the left shows the execution of R code to load a dataset and generate descriptive statistics. The environment pane on the right shows the loaded objects.

```
> # Practical 1: Generating Descriptive Statistics
> # Dataset: airline_flight_delay_analysis.csv
> # -----
> # Install and load required packages
> if(!require(psych)) install.packages("psych")
> library(psych)
> # -----
> # 1. LOAD DATA
> # -----
> df <- read.csv("airline_flight_delay_analysis.csv")
> print("--- Dataset Loaded Successfully ---")
[1] "--- Dataset Loaded Successfully ---"
> # Check column names strictly
> print(colnames(df))
[1] "flight_id" "airline" "origin_airport" "destination_airport"
[5] "scheduled_departure_min" "departure_delay_min" "arrival_delay_min"
[9] "weather_delay_min" "air_traffic_delay_min"
> # -----
> # 2. DESCRIPTIVE STATISTICS
> # -----
> print("--- 1. Descriptive Statistics ---")
[1] "--- 1. Descriptive Statistics ---"
> # A. Using Base R summary()
> # Note: Main column name 'departure_delay' se 'departure_delay_min' kar diya hai
> print("Summary of Departure Delay (in minutes):")
[1] "Summary of Departure Delay (in minutes):"
> summary(df$departure_delay_min)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    -10     66     142     144     221     299
> # -----
> print("Summary of Arrival Delay (in minutes):")
[1] "Summary of Arrival Delay (in minutes):"
> summary(df$arrival_delay_min)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    -10     66     142     144     221     299
```

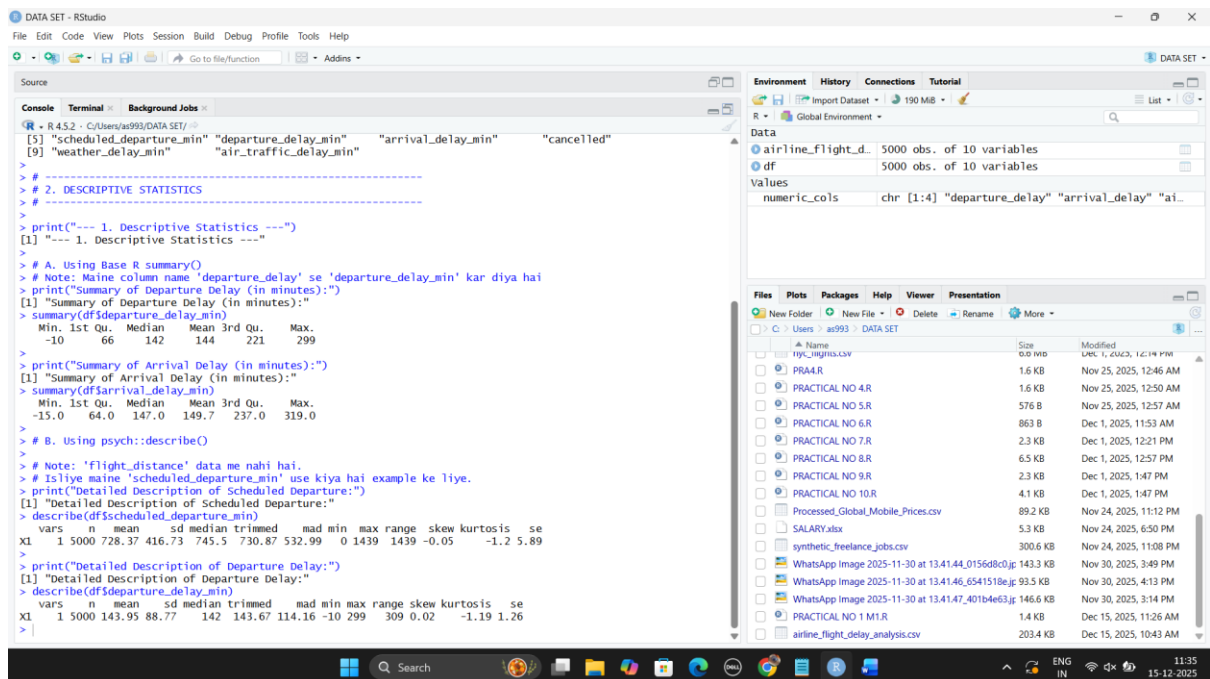
The environment pane on the right shows the following objects:

Object	Class	Attributes
airline_flight_d_	data.frame	5000 obs. of 10 variables
df	data.frame	5000 obs. of 10 variables
numeric_cols	chr	[1:4] "departure_delay" "arrival_delay" "ai_"

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Data Analysis with SAS / SPSS / R



The screenshot displays the RStudio environment with the following components:

- Source Editor:** Contains R code for data analysis, including variable assignment, descriptive statistics, and data frame manipulation.
- Console:** Shows the output of the R code, including summary statistics for 'departure_delay_min' and 'arrival_delay_min'.
- Environment Pane:** Lists the objects in the global environment, including 'airline_flight_d', 'df', and 'numeric_cols'.
- Files Pane:** Displays the file explorer, showing the current directory and various files, including 'PRACTICAL NO 4.R' through 'PRACTICAL NO 10.R'.

```
R • R 4.5.2 • C:/Users/as993/DATA SET /  
File Edit Code View Plots Session Build Debug Profile Tools Help  
Go to file/function  
Source  
Console Terminal Background Jobs  
[5] "scheduled_departure_min" "departure_delay_min" "arrival_delay_min" "cancelled"  
[9] "weather_delay_min" "air_traffic_delay_min"  
>  
> #  
> # 2. DESCRIPTIVE STATISTICS  
> #  
> print("--- 1. Descriptive Statistics ---")  
[1] "--- 1. Descriptive Statistics ---"  
> #  
> # A. Using Base R summary()  
> # Note: Main column name 'departure_delay' se 'departure_delay_min' kar diya hai  
> print("Summary of Departure Delay (in minutes):")  
[1] "Summary of Departure Delay (in minutes):"  
> summary(df$departure_delay_min)  
Min. 1st Qu. Median Mean 3rd Qu. Max.  
-10 66 142 144 221 299  
>  
> print("Summary of Arrival Delay (in minutes):")  
[1] "Summary of Arrival Delay (in minutes):"  
> summary(df$arrival_delay_min)  
Min. 1st Qu. Median Mean 3rd Qu. Max.  
-15.0 64.0 147.0 149.7 237.0 319.0  
>  
> # B. Using psych::describe()  
> # Note: 'Flight_distance' data me nahi hai.  
> # Isliye maine 'scheduled_departure_min' use kiya hai example ke liye.  
> print("Detailed Description of Scheduled Departure:")  
[1] "Detailed Description of Scheduled Departure:"  
> describe(df$scheduled_departure_min)  
vars n mean sd median trimmed mad min max range skew kurtosis se  
x1 1 5000 728.37 416.73 745.5 730.87 532.99 0 1439 1439 -0.05 -1.2 5.89  
>  
> print("Detailed Description of Departure Delay:")  
[1] "Detailed Description of Departure Delay:"  
> describe(df$departure_delay_min)  
vars n mean sd median trimmed mad min max range skew kurtosis se  
x1 1 5000 143.95 88.77 142 143.67 114.16 -10 299 309 0.02 -1.19 1.26  
> |  
Environment History Connections Tutorial  
R • Global Environment  
Data  
airline_flight_d 5000 obs. of 10 variables  
df 5000 obs. of 10 variables  
Values  
numeric_cols chr [1:4] "departure_delay" "arrival_delay" "ai...  
Files Plots Packages Help Viewer Presentation  
New Folder New File Delete Rename More  
C:/Users/as993/DATA SET  
Name Size Modified  
rgc_mgins.csv 0.0 MB Dec 1, 2025, 12:14 PM  
PRA4.R 1.6 KB Nov 25, 2025, 12:46 AM  
PRACTICAL NO 4.R 1.6 KB Nov 25, 2025, 12:50 AM  
PRACTICAL NO 5.R 576 B Nov 25, 2025, 12:57 AM  
PRACTICAL NO 6.R 863 B Dec 1, 2025, 11:53 AM  
PRACTICAL NO 7.R 2.3 KB Dec 1, 2025, 12:21 PM  
PRACTICAL NO 8.R 6.5 KB Dec 1, 2025, 12:57 PM  
PRACTICAL NO 9.R 2.3 KB Dec 1, 2025, 1:47 PM  
PRACTICAL NO 10.R 4.1 KB Dec 1, 2025, 1:47 PM  
Processed_Global_Mobile_Prices.csv 89.2 KB Nov 24, 2025, 11:12 PM  
SALARY.xlsx 5.3 KB Nov 24, 2025, 6:50 PM  
synthetic_freelance_jobs.csv 300.6 KB Nov 24, 2025, 11:08 PM  
WhatsApp Image 2025-11-30 at 13:41:44_015608c0.jpg 143.3 KB Nov 30, 2025, 3:49 PM  
WhatsApp Image 2025-11-30 at 13:41:46_6541518e.jpg 93.5 KB Nov 30, 2025, 4:13 PM  
WhatsApp Image 2025-11-30 at 13:41:47_401b4e63.jpg 146.6 KB Nov 30, 2025, 3:14 PM  
PRACTICAL NO 1 M1.R 1.4 KB Dec 15, 2025, 11:26 AM  
airline_flight_delay_analysis.csv 203.4 KB Dec 15, 2025, 10:43 AM
```

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